Practical-06

Name: Shreya Shedi **PRN: B25ET1193** **Div: SY3 C3**

FAQs

1. What is Queue?
● Queue is a linear data structure that follows the FIFO (First In, First Out) principle.
● Data is inserted from the rear end and deleted from the front end.
● Example: A line of people waiting for a ticket.

2. How is Queue represented?
● A queue can be represented using an array.
● It can also be represented using a linked list.
● Two pointers, front and rear, are generally used to perform queue operations.

3. Enlist types of Queues.
The main types of queues are:
1. Linear Queue
2. Circular Queue
3. Priority Queue
4. Deque (Double-Ended Queue)

4. What is the concept of Circular Queue?
● A circular queue is a queue in which the last position is connected back to the first position.
● When the rear reaches the last position, it can move back to the first position if space is available.
● It helps in efficiently utilizing the empty spaces in a queue.

5. Applications of Queue
● Queues are used in CPU scheduling and printer scheduling.
● They are used in handling requests and managing data in the order of arrival.
● They are also used in simulation of real-life waiting lines.

6. What is the Priority Queue?
● A priority queue is a collection of elements in which each element is assigned a priority.
● An element with higher priority is processed before an element with lower priority.
● If two elements have the same priority, they are processed according to their order of Insertion.

7. What is Deque?
● Deque stands for Double-Ended Queue.
● It is a linear data structure in which elements can be inserted and deleted from both

front and rear ends.
● Elements cannot be inserted or deleted from the middle.